

OHMNI: Hand-Trackled XR Circuit Prototyping with Integrated SPICE Simulation

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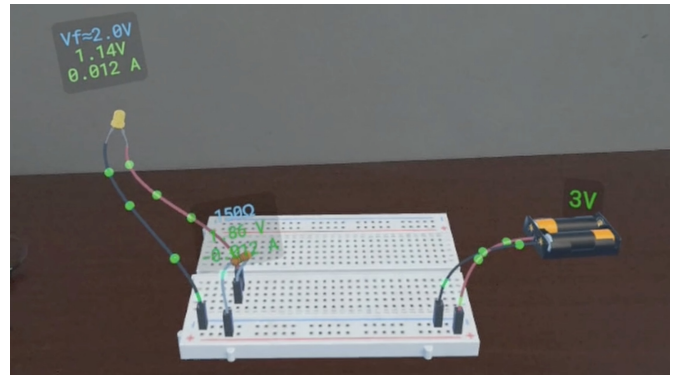
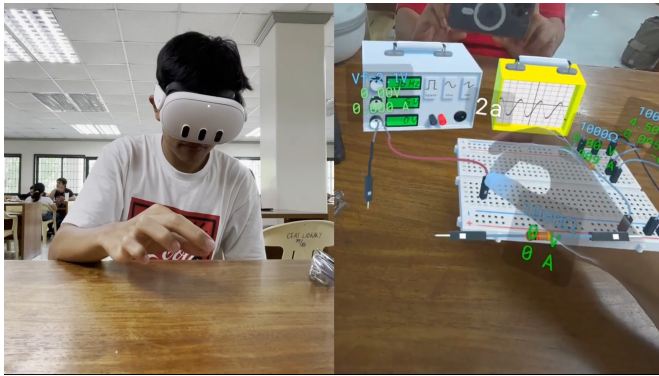


Figure 1: (a) A user building a circuit; right shows the MR view. (b) OHMNI on Meta Quest 3: a virtual LED circuit assembled with labels and electron flow visualization.

Abstract

We introduce OHMNI, an XR application for the Meta Quest 3 that combines hand-tracked 3D circuit assembly with real-time SPICE simulation for immersive circuit prototyping. Benchmarks—including baseline XR comparisons, GPU stress test, GPU vs. CPU load isolation, and a 30-minute stability test—show OHMNI sustains 72fps under moderate load with stable long-session performance.

In a within-subjects study with 20 engineering students, we compared OHMNI with Tinkercad. While OHMNI yielded longer task times (346 s vs. 216 s, $p = 0.009$) and lower SUS scores (62.8 vs. 73.8, $p = 0.021$), 30% of participants performed faster in XR and rated usability as “Good” (73.8). Participants praised enhanced spatial comprehension but noted hand-tracking precision and UI discoverability as challenges. Taken together, the user study and benchmarks position OHMNI as a viable standalone XR circuit prototyping tool.

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1 Introduction

Prototyping is essential in hardware development, enabling early circuit validation before mass production. However, physical breadboarding can be costly and inaccessible, especially in educational or low-resource settings, posing barriers for beginners. 2D simulators like Tinkercad improve reach but lack intuitive spatial interaction. We present OHMNI, an XR system for circuit prototyping on Meta Quest 3 that combines hand-tracked 3D assembly with SPICE simulation, supporting clearer spatial understanding of circuits.

1.1 Problem Statement and Motivation

While XR enables immersive & intuitive interaction, its application in electronics prototyping remains limited—especially on untethered, hand-tracked systems. Existing tools often rely on PC-tethered

or controller-based VR and lack structured usability studies or reproducible performance benchmarks. OHMNI addresses this gap by introducing a hand-tracked interaction technique for circuit assembly on a standalone XR headset, integrated with SPICE simulation and supported by structured user and performance evaluation.

2 Literature Review

SPICE-based tools like LTspice and Multisim have long improved access to electronic prototyping [Nagel and Pederson 1973], but their 2D interfaces constrain spatial reasoning for circuit comprehension. VR simulators such as Short Circuit VR and Project Faraday offer immersive alternatives, with some integrating SPICE-compatible engines like NGspice or SpiceSharp [Bauwens and Ho 2018; Schack 2019]. However, most rely on tethered setups or controllers, limiting portability especially in educational contexts. None also provide structured benchmarks or usability data, leaving gaps in understanding their effectiveness on standalone, hand-tracked XR systems.

3 Methodology

OHMNI was developed in Unity for Meta Quest 3 using the XR Interaction Toolkit (XRI). Users select components from an on-wrist panel menu and spawn them via pinch-drag. Two hand-tracking gestures are supported: pinch to grab components and attach/detach pins, and poke gesture to interact with switches and buttons. When a male and a female pin properly align, they auto-snap and create a node in the netlist; and as soon as a closed loop is detected, the circuit is immediately simulated. All components are custom-modeled in Blender and optimized for mobile rendering to ensure smooth performance.

The simulation core uses SpiceSharp, an open-source SPICE engine implemented in C# on .NET. Upon each connection update, OHMNI performs an Operating Point analysis to compute voltages, currents, and power dissipation, then overlays these values directly on components. AC behavior is approximated by repeated OP solves driven by user-defined frequency and waveform settings, with outputs displayed on a virtual oscilloscope. This architecture delivers responsive, real-time feedback on a standalone XR headset.

3.1 Benchmarking

To systematically characterize OHMNI's performance on consumer XR hardware, we ran four benchmarks on the standalone Meta Quest 3 using OVR Metrics Tool. First, a baseline test compares OHMNI against (1) a blank Unity XR Interaction Toolkit template and (2) Meta's First Hand demo to quantify system overhead. Second, a 3D rendering test scales resistor models from 3 to 60 to isolate GPU-bound costs. Third, an AC simulation test contrasts full rendering with a zero visuals "simulation-only" mode to separate CPU load from GPU work. Finally, a 30-minute stability test logs frame rate, thermal, and memory metrics to reveal drifts. Together, these benchmarks isolate graphical and computational bottlenecks critical to maintaining the 72fps target in an immersive, hand-tracked circuit-prototyping tool [Meta Platforms, Inc. 2024].

3.2 User Study

A within-subjects usability study was conducted with 20 Electrical Engineering students from the University of the Philippines Los

Baños. Each completed the same task of connecting an AC waveform generator to a resistor and LED, then adjusting parameters to make it pulse—using both OHMNI (Meta Quest 3 with hand-tracking) and Tinkercad (MacBook Air M2). Platform order was counterbalanced. Task completion time, SUS scores, and informal feedback were recorded to compare usability and interaction differences. We used paired t -tests with $\alpha = .05$. All participants provided informed consent.

4 Results, Discussion, and Future Work

While OHMNI maintained stable performance (72fps) under moderate loads, comparable to blank XRI template and Meta's First Hand, usability tests showed it lagged behind Tinkercad in task time (346s vs. 216s, $p = 0.009$) and SUS scores (62.8 "OK" vs. 73.8 "Good", $p = 0.021$) [Bangor et al. 2009]. For this, we hypothesize three contributors: (1) hand-tracking inaccuracy causing repeated attach attempts, (2) slower component alignment in 3D than simple 2D dragging, and (3) UI discoverability. Yet, its spatial interactions offered clear pedagogical value: many participants reported improved circuit understanding through embodied 3D manipulation, echoing findings that physical interaction enhances conceptual learning [Wilson 2002]. Performance bottlenecks emerged with complex circuits, where rendering and hand-tracking limitations impaired responsiveness, underscoring how frame stability critically affects XR interaction fidelity. To address this, future versions will shift to SpiceSharp's native AC solver to reduce CPU load, and apply GPU instancing with adaptive LOD for better rendering. Usability will be improved via timed snap-to-pin zones, stylus/controller support, and progressive onboarding to ease learning. Planned usability improvements are summarized in Table 1.

Table 1: Planned improvements derived from study findings.

Limitation	Planned improvement
Depth perception	Plane detection and shadows on detected surfaces for depth cues
Hand-tracking imprecision	Time-gated snap-to-pin zones for connecting pins; optional stylus/controller
Poor UI discoverability	Contextual toast hints and ghost-hand gesture cues for guidance
Visual clutter (complex circuits)	Hybrid 2D schematic overlay with a toggle between 2D/3D views
Steep learning curve	Progressive onboarding tutorials with stepwise tasks
Weak action feedback	Audio/visual confirmations

As XR platforms evolve, OHMNI establishes a promising foundation for immersive circuit prototyping. With continued optimization, it can help redefine electronics education and rapid prototyping through intuitive spatial interfaces and accessible simulation.

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